1. **Problem Statement and Motivation**
   - **Real-World Business Problem:** Shared costs among friends can be difficult to manage and cause “mental load.” The process of calculating how much each person owes for things like dinners, trips, and household utility bills is often prone to human errors. This is especially apparent when dealing with tax, tips, and itemized splits. In addition to the math, the social burden of being responsible for reminding people about their unpaid debt causes awkwardness and can lead to financial loss.
   - **Target Users:** College students, travel groups, roommates
   - **Why an Agentic AI Solution:** An Agentic AI Solution is specifically appropriate here because it has many advantages over a traditional static/rule-based automated app. The agentic approach offers an autonomous, goal-oriented AI system capable of using real-world, unclean data. Traditional apps are typically designed such that humans need to manually specify each action that will be performed on a receipt, however, this can create "friction fatigue" leading to abandonment of the use of the app. An agentic app provides a workflow with multiple agents working together to perceive unstructured receipts through multi-modal perception, and to perform complex, non-linear splits (e.g., split costs for food at a table where some individuals did not eat from all items ordered; split taxes based upon different rates per item purchased). Additionally, the use of an autonomous agent to collect payments removes the potential for social awkwardness and personal conflict associated with humans collecting payments from other humans, and the potential for the mediator to be seen as biased/subjective.

2. **Use Cases & Scenarios**
    - **Complicated Group Dinner**
        i. **User:** User takes a photo of a $200 receipt for a table of six and tells the agent “Mark and I split the nachos; everyone else split the pizza.”
        ii. **Agent Tasks:** The Accountant (vision) agent digitizes the receipt into specific line items. The Divider (logic) agent calculates the exact subtotal, tax and tip per person. The Collector (communication) agent automatically generates unique Venmo/Zelle “Pay” links for each person and pings them with the breakdown so they can pay instantly.
    - **Weekend Trip**
        i. **User:** At the end of a weekend trip, the group uploader sends 10 different receipts (gas, Airbnb, groceries) totaling $1,300.
        ii. **Agent Tasks:** The Accountant (vision) agent digitizes all the receipts, categorizes the spending, and identifies which members paid for each initial transaction. The Divider (logic) agent performs a net settlement calculation allowing members who overpaid to be directly reimbursed by those who underpaid in the least amount of transactions possible. The Collector (communication) agent sends a message to each person “Trip Summary: You spent $40 but your share was $150. Please pay [Name] $110 here: [Venmo Link].
3. **Initial System Design**
    - **Preliminary List of Agents and their Roles**

    i. **Accountant (Data Fetcher):** A vision-enabled agent that digitizes receipt line items, taxes, etc.
    ii. **Divider (Logic):** Maps users to specific expenses and calculates tax/tip and net settlements.
    iii. **Collector (Communication):** Constructs Venmo/Zelle links and triggers automated notifications for payments
    iv. **Auditor:** Verifies mathematical accuracy by balancing individual splits against grand total.
- **Expected Tools/APIs/Data Sources:** GPT-4o for multi-modal reasoning, Twilio API for sending notifications to individuals, local image files,

4. **Feasibility & Risks**
- **Potential Challenges & Constraints:** Possibility that an LLM will misrepresent extracted receipt data when utilizing low quality receipts, and the likelihood of failure with an extraction tool when the API used to deliver external services has an outage. Also API rate limits for higher resolution image capture that the system must address in order to be able to operate successfully including the cost per message to send an SMS message. Finally, another major constraint is user data privacy (financial, phone numbers, etc).
- **Timeline for Project Execution:**
    i. Develop the front-end interface and receipt upload functionality (by 3/11)
    ii. Implement the multi-agent workflow connecting vision, logic and communication (by 3/25)
    iii. Run pilot tests and refine agent accuracy (by 3/30)